Malaria prevalence and child mortality — Methods & Results

Malaria prevalence and child mortality — Methods & Results summary

Three analyses relating *P. falciparum* prevalence to all-cause under-5 mortality in sub-Saharan Africa. All figures/tables below are reproduced by `run_all.R` into `results/`.

Data sources: MAP $PfPR_{2\text{-}10}$ raster (2024); IHME/GBD under-5 malaria deaths (2025); UN IGME all-cause child mortality + World Bank live births, **GDP per capita** (`NY.GDP.PCAP.CD`) and **DTP3** (`SH.IMM.IDPT`); DHS/MIS microdata (PR + birth-history recodes) across 23 SSA countries.

Component 1 — Country level

Methods. Malaria's share of all-cause under-5 deaths = IHME under-5 malaria deaths ÷ IGME all-cause under-5 deaths (U5MR/1000 × live births), computed for **all under-5** and **neonatal-excluded** (denominator = 1mo–5y = U5MR - neonatal). Exposure = population-weighted national **$PfPR_{2\text{-}10}$** (MAP 2024). Linear model `share ~ PfPR + log(GDP p.c.) + DTP3`; outlier countries flagged by studentized residual from the prevalence-only fit.

Results (42 SSA countries). Malaria's share of child deaths rises strongly with prevalence: ≈ **+0.9 percentage-points of the all-U5 share (≈ +1.3 pp of the 1mo–5y share) per +1 $PfPR_{2\text{-}10}$ point** (adjusted for GDP and DTP3; adjusted $R^2 \approx 0.5$–0.6). **Burundi** is a striking outlier (studentized residual ≈ 8) — malaria a far higher share of child deaths than prevalence predicts — followed by Uganda, Rwanda and Ghana. GDP per capita is not significant once DTP3 is included.

Tables: `results/component1_model_coefficients.csv`, `component1_outliers.csv`.

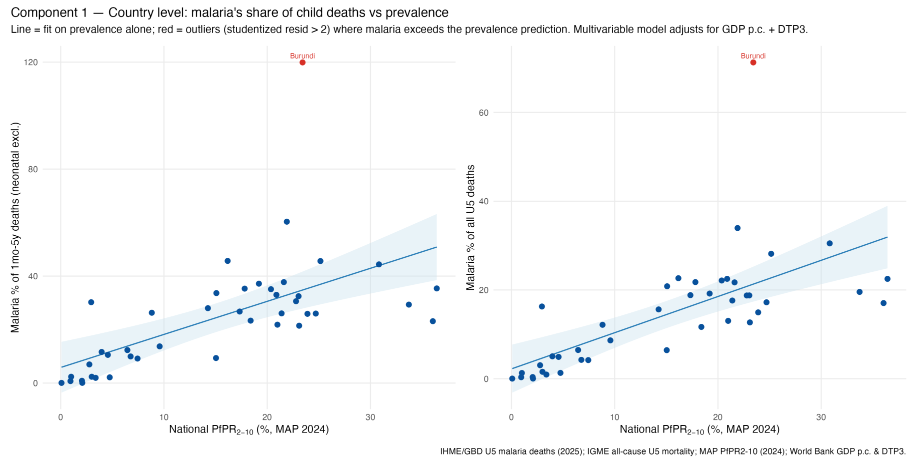

Figure 1: Component 1: malaria share of child deaths vs national PfPR

Component 2 — DHS subnational + national

Methods. One observation per **survey-region** across all usable DHS/MIS surveys (malaria biomarker + birth history). All-cause **U5MR** and **1mo–5y** mortality from the birth histories via `DHS.rates::chmort`. Exposure = child malaria prevalence put on a common footing: **microscopy-equivalent** (RDT to microscopy via the Component-3 factor where microscopy is missing), then **age-standardised 0.5–5y to PfPR$_{2\text{-}10}$ (2–10y)** with `malariaAtlas::convertPrevalence` (Smith et al. 2007 age–prevalence model). Two specifications:

1. **Primary — log-rate linear mixed model:** `log(mortality) ~ PfPR2-10 + DTP3 + log(GDP) + %urban + year + (1 + PfPR2-10 || country)`.
2. **Count — negative-binomial GAM (`mgcv`):** region under-5 deaths with a `log(exposure)` offset, a smooth `s(year_c)`, and country random intercept + slope.

Child stunting was evaluated as a further confounder but **excluded from the primary model**: five MIS-type surveys (including Liberia's only survey) collected no anthropometry, so adding it dropped ~11% of survey-regions and one country while leaving the malaria coefficient essentially unchanged.

Results (600 survey-regions, 44 surveys, 23 countries). Higher child malaria prevalence predicts higher child mortality, adjusted for all covariates:

	LMM (log-rate)	GAM (nb count)
U5MR, per +10 PfPR$_{2\text{-}10}$ pts	**+7.0%** (95% CI +3.2, +10.9)	+6.9% (+3.5, +10.4)

	LMM (log-rate)	GAM (nb count)
1mo–5y, per +10 pts	**+11.3%** (+5.8, +17.0)	+11.0% (+5.8, +16.5)

The effect is **positive in every country** (country random slopes; figure below) and covariates behave as expected — DTP3 and % urban protective, a ~3%/yr secular decline, GDP ~null after the others. **Sensitivity** restricting to mid-transmission regions ($PfPR_{2\text{-}10}$ 5–50%, 354 regions) attenuates the effect (LMM: U5MR +6.2%, 1mo–5y +8.4%; GAM: +5.6% / +8.7%) — positive but weaker/less precise over the narrower range.

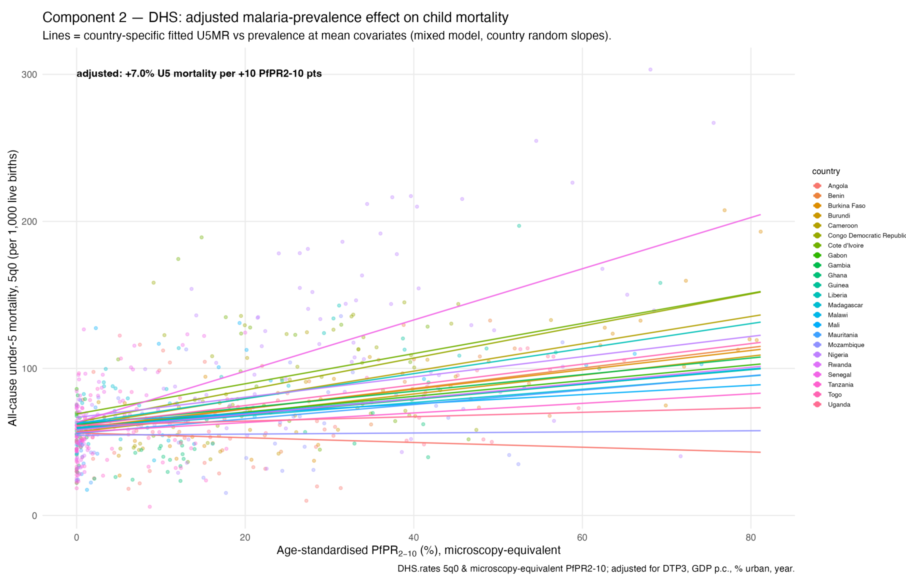

Figure 2: Component 2: adjusted country-specific prevalence–mortality slopes

Exposure-response shape. Replacing linear prevalence with a low-df smooth `s(PfPR~2-10~, k=4)`: **U5MR is linear** across the range (edf = 1.0 — a constant proportional effect), while **1mo–5y shows mild curvature** (edf ≈ 2.1 — a smooth, decelerating rise, steeper at low prevalence). Removing neonatal deaths from the denominator uncovers the modest nonlinearity the all-U5 relationship hides.

Tables: `results/component2_model_coefficients.csv`, `component2_count_model_coefficients.csv` (`sample` = full / pfpr_5_50), `component2_country_slopes.csv`, `component2_exposure_response_edf.csv`. Sensitivity figure: `component2_sensitivity_5to50.png`.

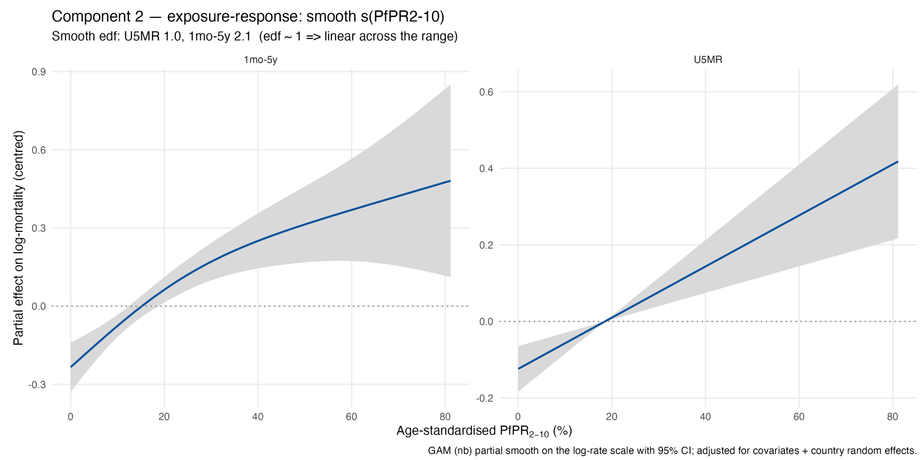

Figure 3: Component 2: smooth exposure-response s(PfPR2-10)

Component 3 — RDT vs microscopy (supporting)

Methods. Across all DHS/MIS survey-regions with both tests, regress microscopy prevalence on RDT prevalence to a consensus conversion factor, used by Component 2 to impute microscopy where only RDT exists.

Results. RDTs over-detect (persistent HRP2 antigen), so microscopy runs lower: **microscopy ≈ 0.74 × RDT** (Pearson r = 0.90, 431 survey-regions).

Table: `results/rdt_microscopy_conversion.csv`.

Cross-component prediction — total child mortality, 10% to 30% $PfPR_{2\text{-}10}$

Holding covariates at their means and anchoring both models to the same baseline at 10% PfPR (within each outcome), the predicted change in **total** all-cause child mortality as prevalence rises from 10% to 30%:

Outcome	baseline @10%	Component 1 (share, non-malaria fixed)	Component 2 (direct model)
All under-5 (U5MR)	64.3 / 1,000	**+23.8%** to 79.6	**+14.4%** to 73.6
1mo–5y (neonatal excl.)	38.3 / 1,000	**+48.3%** to 56.8	**+23.8%** to 47.4

Component 2 (direct total-mortality model) gives the empirical net effect; Component 1 (malaria's share, assuming non-malaria mortality is fixed) gives the upper "malaria-as-pure-addition" bound[1]. Effects are larger for 1mo–5y because the malaria-irrelevant neonatal floor is removed. The 95% CIs (shaded) overlap, so the two independent routes are consistent. *(1mo–5y is nonlinear — see the exposure-response above — so its log-linear prediction is an approximation over this span.)*

Malaria attributable fraction of child deaths

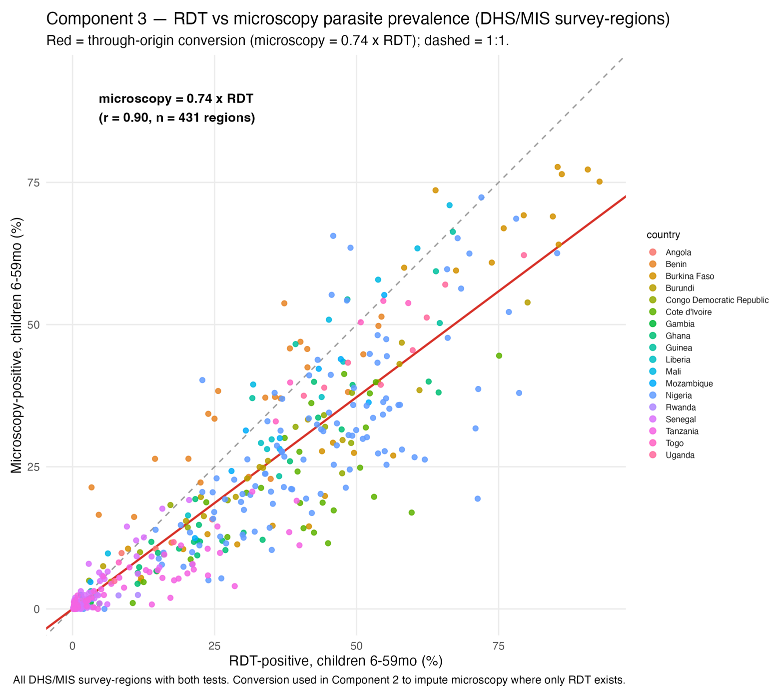

Figure 4: Component 3: RDT vs microscopy

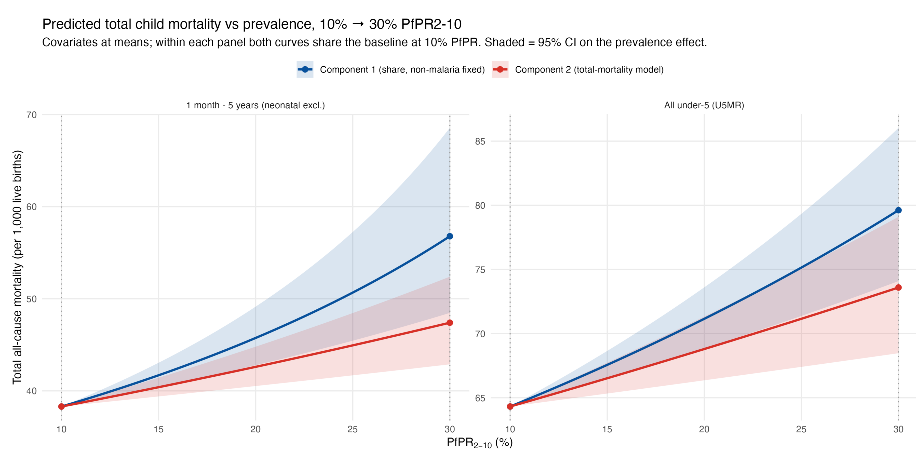

Figure 5: Predicted total child mortality vs prevalence, 10% to 30% PfPR2-10

$N \times \mathrm{AF}(p)$. Shown for both outcomes as the **linear** closed form (LMM coefficient) and the **spline** (low-df nb-GAM `s(PfPR, k=4)`); the neonatal-excluded denominator uses N = U5MR - NNMR.

$\mathrm{PfPR}_{2\text{-}10}$	All under-5, linear	All under-5, spline	1mo–5y, linear	1mo–5y, spline
10%	6.5%	6.5%	10.1%	14.6%
20%	12.6%	12.5%	19.2%	25.7%
30%	18.3%	18.2%	27.4%	33.3%
40%	23.6%	23.5%	34.7%	38.4%
50%	28.6%	28.4%	41.3%	42.2%

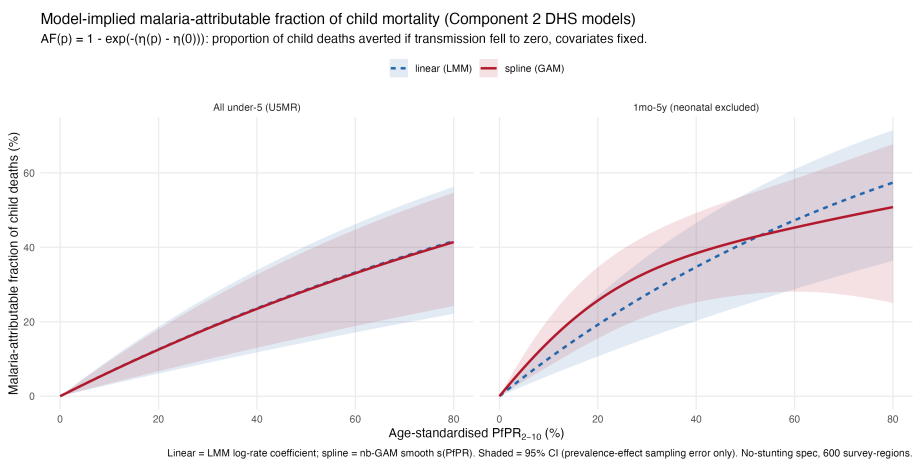

Figure 6: Malaria-attributable fraction of child deaths vs PfPR2-10

For **U5MR** the linear and spline curves coincide (exposure-response is linear, edf $\approx$ 1.0). For **1mo–5y** the smooth (edf $\approx$ 2.1) rises a little faster than the linear form at low–mid prevalence then flattens, the two converging near ~40% by 50% PfPR — a mild, decelerating nonlinearity rather than the over-flexible wiggle a high-df spline produced. Removing neonatal deaths (largely non-malarial) still roughly doubles malaria's share versus all-U5. These are counterfactual (vs zero-transmission) fractions assuming the adjusted association is causal; the CIs (shaded) reflect only the prevalence-effect sampling error. Table: `results/attributable_fraction.csv`.

Comparing the two methods

Both components estimate the same quantity — malaria as a **% of child deaths** — so they can be overlaid on shared axes. Component 1 is the country-level share (points = IHME ÷ IGME per country; line = fitted `share ~ PfPR + log GDP + DTP3` at mean covariates); Component 2 is the DHS attributable-fraction spline. Drawn over Component 1's national-prevalence range (to ~40%).

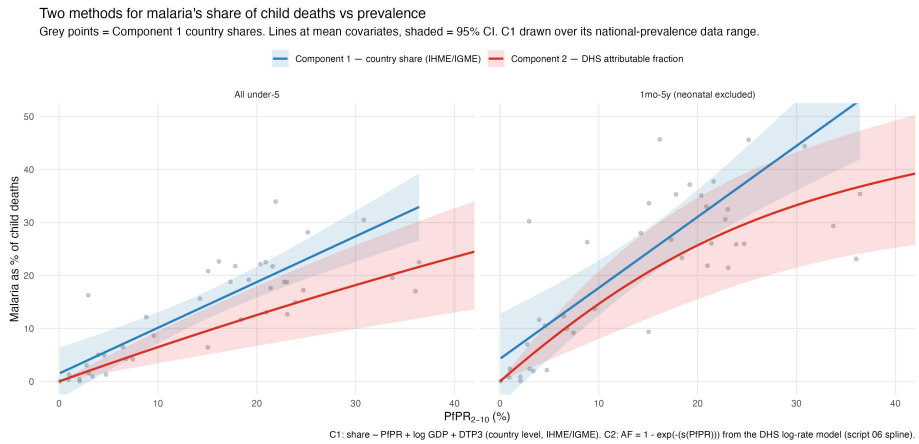

Figure 7: Method comparison: Component 1 share vs Component 2 attributable fraction

The routes **broadly agree** and their 95% bands overlap throughout. For **all under-5** the two track the country cloud closely, Component 2 running slightly below Component 1. For **1mo–5y** both rise more steeply once neonatal deaths are removed; they coincide at low–mid transmission and diverge at the top of the range, where Component 1 extrapolates roughly linearly while Component 2 begins to plateau. That two independent data sources (IHME/IGME country totals vs DHS survey mortality) and two different models land on a similar prevalence–share relationship is the main cross-check of the work.

National malaria-burden estimates ($PfPR_{2\text{-}10} > 10\%$)

Applying both methods to a common denominator — all-cause child deaths = mortality rate × live births (IGME/WB) — gives the number of malaria-attributable child deaths per country. **Component 1** multiplies by the fitted malaria *share* (`share ~ PfPR + log GDP + DTP3`); **Component 2** by the attributable fraction `AF = 1 - exp(-$\beta$·PfPR/10)` from the DHS log-rate model. Both are shown for **all under-5** and **1mo–5y (neonatal-excluded)**, with the outcome-matched coefficients. Only the malaria coefficient carries uncertainty (prevalence, GDP, DTP3, births, rates fixed); because that coefficient is shared across countries, a pooled-total CI is the total re-evaluated at the coefficient's bounds. A national/subnational check for Nigeria and DR Congo (below) showed aggregating admin-1 estimates moves the total by <1%, so national PfPR is the representative scale.

Malaria-attributable child deaths for the **25 countries** with national $PfPR_{2\text{-}10}$ > 10% (point estimates; sorted by Component 2 all-U5). C1 = fitted share, C2 = attributable fraction:

Country	PfPR	C1 (U5)	C2 (U5)	C1 (1mo–5y)	C2 (1mo–5y)	IHME
Nigeria	24.7%	178,352	133,962	189,730	134,093	150,493
DR Congo	36.0%	125,535	86,371	146,432	94,214	68,300
Niger	17.3%	26,496	13,657	31,259	14,716	23,329
Uganda	21.9%	21,246	11,502	19,985	9,826	28,474
Mozambique	23.9%	17,193	11,324	16,013	9,892	11,388
Cameroon	25.2%	13,727	9,807	13,872	9,137	17,718
Angola	20.4%	8,394	8,841	8,550	8,489	15,266
Côte d'Ivoire	21.6%	11,184	8,806	10,665	7,695	14,113
Chad	15.0%	10,901	8,232	12,325	8,704	5,498
Mali	17.8%	13,424	7,988	13,846	7,533	15,377
Benin	36.4%	10,138	7,854	10,176	7,388	8,123
Burkina Faso	20.9%	13,717	7,254	15,672	7,529	12,439
Guinea	21.4%	6,608	6,077	7,160	6,247	7,979
South Sudan	23.1%	7,040	4,861	6,778	4,357	4,282
Central African Rep.	33.7%	5,179	4,487	5,230	4,446	4,320
Sierra Leone	30.8%	8,044	4,414	9,076	4,533	7,181
Malawi	19.2%	8,494	3,991	7,334	3,145	6,316
Ghana	16.2%	5,914	3,305	5,074	2,515	7,254
Burundi	23.4%	7,093	3,207	6,937	2,887	15,659
Zambia	14.3%	6,196	3,091	5,934	2,655	5,272
Togo	21.0%	3,570	1,993	3,575	1,810	1,972
Liberia	15.1%	2,656	1,432	2,774	1,363	3,088
Congo-Brazzaville	23.0%	1,439	1,083	1,374	948	1,414
Equatorial Guinea	22.8%	456	549	458	511	724
Gabon	18.4%	118	262	96	200	262
TOTAL (25)	—	**513,116**	**354,352**	**550,323**	**354,836**	**436,242**

Pooled-total 95% CIs (coefficient-only): all-U5 — C1 [357,649–668,584], C2

[172,582–519,653]; 1mo–5y — C1 [367,934–732,712], C2 [200,714–489,777]. All-cause denominators (2,292,071 U5; 1,507,890 post-neonatal) and every cell's CI are in `results/country_malaria_deaths_gt10.csv`.

The IHME reference (436k, all-U5) sits between the two model routes (C1 ~18% above, C2 ~19% below). Component 2 is internally consistent across denominators (355k either way — the higher post-neonatal fraction offsets the smaller base); Component 1's two share models are fit separately and differ by ~7%. Nigeria + DR Congo alone are ~60% of the burden. Per-country estimates (all columns, with CIs) are in `results/country_malaria_deaths_gt10.csv`; the largest single-country divergences from IHME are **Burundi** (IHME far higher than either model) and **Chad** (lower).

National vs subnational (admin-1) check — all under-5:

	National	Subnational (Σ admin-1)
Nigeria — Component 1	178,352	178,350
Nigeria — Component 2	133,962	133,488
DR Congo — Component 1	125,535	125,535
DR Congo — Component 2	86,371	85,696

Component 1 is unchanged (its share model is linear in prevalence); Component 2 is marginally lower subnationally (the AF curve is concave, so disaggregating high/low-prevalence units loses a little via Jensen's inequality) — but within-country prevalence heterogeneity is small, so the effect is <1%. Scripts: `R/07_ng_drc_malaria_deaths.R` (`results/ng_drc_malaria_deaths.csv`), `R/08_country_malaria_deaths.R` (`results/country_malaria_deaths_gt10.csv`).

RCT triangulation — does the relationship hold in randomised trials?

As an external check, we asked whether the model-implied prevalence to mortality relationship holds in **randomised** insecticide-treated-net (ITN) trials. Using the Pryce et al. (2018) Cochrane ITN review, we took every sub-Saharan **cluster-RCT** that measured **both** a change in child parasite prevalence and all-cause under-5 mortality, extracting the numbers from the **original publications**: **D'Alessandro 1995** (Gambia, 5 zones), **Habluetzel 1997/99** (Burkina Faso), **Phillips-Howard 2003 / ter Kuile** (western Kenya) and **Nevill 1996 / Snow** (Kilifi). Binka 1996 measured no parasite prevalence (excluded); Snow 1987 is individually randomised (excluded).

Method. Each arm's parasite prevalence is age-standardised to $PfPR_{2\text{-}10}$, and **Component 2** predicts the mortality reduction from the observed prevalence reduction by evaluating its **nonlinear spline at both the control and intervention prevalence levels**. (Component 1's linear share model is unusable per-arm here — extrapolated to these prevalences it implies shares > 100%.)

Trial (zone)	Baseline $PfPR_{2\text{-}10}$	Prevalence reduction (pts)	Observed U5 ↓	Component 2 predicted
D'Alessandro z1	37%	9	62%	7%
D'Alessandro z2	34%	10	51%	10%
D'Alessandro z3	26%	10	51%	11%
D'Alessandro z4	54%	10	5%	6%
D'Alessandro z5	46%	-27 (rose)	-20%	-16%
Habluetzel	95%	8	15%	4%
Phillips-Howard	77%	14	15%	7%
Nevill	39%	20	30%	18%

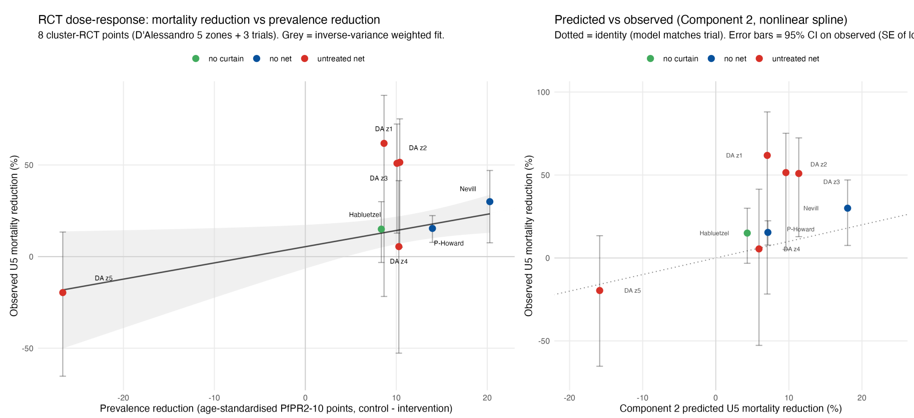

Figure 8: RCT triangulation: dose-response and Component 2 predicted vs observed

Findings. The trials show the expected **positive dose-response** — bigger prevalence reductions go with bigger mortality reductions — at an inverse-variance-weighted ≈ **8.6% fewer U5 deaths per 10 $PfPR_{2\text{-}10}$ points removed**, close to Component 2's own ≈ **11.3% per 10 points**. Component 2 correctly predicts the **negative control** (D'Alessandro zone 5, where the programme failed: prevalence *rose* and mortality *rose*) and the low-effect points (zone 4, Habluetzel, Phillips-Howard). It **under-predicts** the largest observed reductions (D'Alessandro zones 1–3, Nevill), but those rest on few deaths, infant-age or near-saturated prevalence, and their **95% CIs overlap** the model line. The experimental evidence is thus broadly **consistent** with the observational Component 2 relationship. Exploratory (8 heterogeneous points: curtains, an untreated-net comparator, spillover, infant-age; D'Alessandro-zone CIs are Poisson, un-clustered). Data: `results/triangulation_rct_data.csv`; script `R/10_triangulation_rct.R`.

Caveats

- **Component 1 is ecological** (country-level); **Component 2** adjusts for the main confounders but is observational, and region-level U5MR from a single survey is noisy.
- MAP $PfPR_{2\text{-}10}$ is 2024 vs the 2025 IHME export (1-year offset).
- IHME (GBD) and WHO differ substantially on malaria mortality; Component 1's numerator is IHME.
- DHS prevalence is age-standardised to $PfPR_{2\text{-}10}$ via a model (Smith 2007), and RDT is converted to a microscopy footing; both are documented approximations.